

More Than a Consensus? A Social Relations Model Analysis of Status

Abstract

Decades of research have advanced collective knowledge of the nature, antecedents, and consequences of status. This work has predominantly conceptualized status as a consensus that emerges within social groups. Here, we argue that status encompasses all evaluations of respect and admiration, whether or not they are reflected in a group's consensus. We examine dyadic status effects: components of status evaluations that are unique to particular perceiver-target pairs. Using the social relations model, we demonstrate that a significant amount of variance in status evaluations can be attributed to dyadic status effects in three samples drawn from different geographic and institutional contexts ($N = 12,464$ status observations). We then theorize and show that dyadic status effects are greater as the status distance between two members of a dyad increases.

Keywords: status, social relations model, dyadic data, status distance

Status is the relative extent to which individuals are respected and admired in a social group (Magee & Galinsky, 2008). Status has drawn considerable scholarly attention because it shapes important individual and group outcomes (Berger et al., 1972; Blau, 1964; Magee & Galinsky, 2008). Moreover, status structures social interaction, for instance by shaping the opportunities an individual receives (and accepts) to participate and exert influence over others (Anderson & Brown, 2010; Magee & Galinsky, 2008). Consequently, understanding the drivers of status has been an enduring concern for scholars.

Scholars have proposed that individuals are conferred status based on the perceived instrumental value they provide to others in a social group. Individuals deemed the most competent and prosocially motivated are rewarded with the highest status (Anderson et al., 2015; Berger et al., 1972; Henrich & Gil-White, 2001; Ridgeway & Berger, 1986). This has led scholars to examine numerous individual-level antecedents of status. Indeed, prominent theories, notably expectation states theory and status characteristics theory (Berger et al., 1972), assert that certain attributes and behaviors elicit broad agreement in terms of the status they generate for particular target-individuals. This literature has demonstrated that attributes such as demographic characteristics (e.g., gender, ethnicity, and age; Bunderson, 2003; Ridgeway & Berger, 1986), personality traits (e.g., extraversion; Anderson et al., 2001; Grosz et al., 2023; Judge et al., 2002), and behaviors (e.g., overconfidence; Anderson et al., 2012) play a central role in generating status for the same individuals across different social groups.

However, in the present research, we argue that this individual-level approach to studying the antecedents of status overlooks another important source of variation in status evaluations. Dyadic status effects are components of status evaluations that are unique to particular perceiver-target pairs after accounting for perceiver and target effects. These effects develop when pairs of

individuals—perceivers and targets—interact in the dyadic relationships that are nested within their larger social groups. A small body of status research has begun to examine dyadic status effects (e.g., Joshi & Knight, 2015); however, this literature lags behind scholarship in other areas of interpersonal perception, which has shown that dyad-specific factors explain significant variation in interpersonal judgments (e.g., de Jong et al., 2007; Jones & Shah, 2016). Although literature on dyadic status effects remains limited (for an exception, see Joshi & Knight, 2015), nascent work has highlighted that there is often substantial dissensus in status evaluations (Kilduff et al., 2016; Tan & Rider, 2023; Yu et al., 2023; Yu & Kilduff, 2020). Dyadic status effects are one statistical component of dissensus; identifying their antecedents may therefore help explain why dissensus develops.

Thus, in the present research, we extend individual-level accounts of status by examining dyad-specific variation. Specifically, we investigate two core research questions. First, to what extent do status evaluations in groups reflect dyadic status effects? To answer this question, we employ social relations model analyses (SRM; Kenny & La Voie, 1984) to decompose variance in status evaluations into perceiver, target, and dyadic components. Although prior research on status has employed SRM, this work has focused on verifying that a statistical consensus emerges, which justifies studying individual-level antecedents of status evaluations (e.g., DesJardins et al., 2015). Additionally, to the best of our knowledge, prior SRM analyses of status have not separated dyadic variance from residual error, which precludes reliably estimating dyadic status effects (Schönbrodt et al., 2012). Consequently, we use SRM to comprehensively analyze status evaluations from three samples drawn from different geographic and institutional contexts ($N = 12,464$ status observations). In each sample, we find that a significant amount of

variance in status evaluations is attributable to dyadic status effects, indicating their clear importance for understanding the emergence of status.

Second, we ask whether some dyads generate larger dyadic status effects than others. We approach this question by thinking about how an individual's own standing relative to a dyadic peer may drive or suppress the emergence of dyadic status effects. We argue that perceivers generate greater dyadic status effects for targets whose status in the group consensus is farther from their own—that is, targets at a greater status distance. In contrast, we argue that perceivers generate lower dyadic effects for targets whose status is more similar to their own. We find support for this hypothesis in each of our empirical samples.

We make several important contributions to the status literature through the present research. First, we contribute to a deeper understanding of status emergence by highlighting the importance of dyadic status effects, thereby advancing knowledge about an underexplored component of status evaluations. We demonstrate that dyadic status effects are critical to explaining and understanding status, using SRM, which provides clear impetus for future research. Second, we generate knowledge about where these dyadic status effects are most likely to emerge from. Specifically, we suggest that greater dyadic status effects emerge when members of dyads are at greater, rather than smaller, status distances from one another. Lastly, our work contributes greater understanding as to why groups often reach relatively weak consensus over one another's positions in their status hierarchies (Anderson & Brown, 2010; Yu et al., 2023; Yu & Kilduff, 2020), which is crucial because a lack of consensus undermines performance (Bendersky & Hays, 2012; Kilduff et al., 2016).

Theory and Hypotheses

Social groups confer status based on the perceived instrumental value that each individual provides (Anderson et al., 2015; Berger et al., 1972; Ridgeway & Berger, 1986). Perceived instrumental value comprises the extent to which individuals are seen to be competent in tasks relevant to the group's success, and whether individuals are perceived to demonstrate the willingness to advance the interests of the social group (Bai et al., 2020; Flynn, 2003; Hardy & Van Vugt, 2006; Ridgeway, 1978; Willer, 2009). Those seen as most able and willing to contribute are deemed most worthy of status.

Competence comprises the skills and expertise needed to address a group's challenges, including the ability to contribute useful ideas and information (Bai et al., 2020). Competence perceptions are partly influenced by status characteristics—attributes that have acquired status value through widespread association with social worth and competence (Bunderson, 2003; Correll & Ridgeway, 2006). These characteristics may include gender, ethnicity, age (Bunderson, 2003; Ridgeway & Berger, 1986), and personality traits (Anderson et al., 2001; Grosz et al., 2024; Judge et al., 2002). Competence is also signaled by behaviors: for instance, overconfident individuals signal their competence to others and subsequently acquire status (Anderson et al., 2012), while displays of authentic pride (Cheng et al., 2010) and the effective use of humor (Bitterly et al., 2017) have also been shown to drive perceived competence. Thus, scholars have identified a diverse set of attributes and behaviors that drive the extent to which individuals are perceived to be competent and motivated to advance the interests of their groups. These individual-level attributes and behaviors have so far served as the foundation for understanding the antecedents of status.

However, perceptions of competence—and therefore of status—are highly subjective, particularly in complex task environments where there are few objective markers of performance (Foschi, 2000; Inesi & Cable, 2015; Kunda, 1990). For instance, Kunda (1990) demonstrated that perceivers apply and weigh different criteria when evaluating targets' competence, and they are often motivated by the desire to confirm existing beliefs. Nickerson (1998) reviewed evidence that confirmation bias shapes how perceivers interpret performance, leading to diverging performance evaluations across different dyads. Perceivers' own attributes also shape competence judgments. In a military boot-camp study, Kalish and Luria (2021) found that participants used different criteria to attribute emergent leadership to peers: some emphasized physical ability, whereas others emphasized cognitive ability. These criteria aligned with perceivers' own attributes: physically capable perceivers placed more weight on physical ability, whereas cognitively capable perceivers placed more weight on cognitive ability. Importantly, these attributions aligned with perceivers' own status characteristics, whereby those perceivers with strong physical abilities were more likely to attribute leadership to their physically capable peers, while those with strong cognitive abilities did so based on peers' cognitive abilities. Together, this evidence shows that competence attributions are not uniform across perceivers but may instead be shaped by dyadic effects that emerge specifically between combinations of perceivers and targets.

A hypothetical example illustrates how these dynamics might play out in a group setting. Imagine a project team working on a task. Within the group, one individual (the target) might be seen as highly competent by one particular individual (the first perceiver). Perhaps both the target and the first perceiver share similar levels of extraversion, and this produces a dyadic status effect such that the first perceiver confers especially high status on the target. However,

perhaps another individual (the second perceiver) evaluates the same target as being significantly less competent. This second perceiver may be less extraverted than the target, meaning that the dyadic status effect produced in the first dyad is not repeated in the second. Thus, while the first perceiver attributes high competence to the target, the second does not. In turn, this means that the first perceiver confers high status on the target, while the second perceiver confers lower status on the same target. As these, and other, dyadic status effects proliferate across dyads in a social group, a substantial amount of variance in status begins to emerge due to factors that are only explainable at the dyadic level. Put differently, dyadic status effects are not explained by the main effects of individual attributes or behaviors; they instead reflect particular combinations or interactions of perceivers and targets.

In this way, the subjective nature of competence and status evaluations mirrors perceptions of other interpersonal constructs, which are known to vary across dyads. Research has shown, for instance, that individuals build social affinities and form trust judgments based on dyadic effects, rather than solely individual-level attributes or behaviors (de Jong et al., 2007; Jones & Shah, 2016; Joshi & Knight, 2015). We predict that status evaluations, much the same as liking and trust, is highly influenced by dyadic effects.

Which Dyads Show Greater Dyadic Status Effects?

In addition to theorizing that status is conferred, in large part, through dyadic status effects, we further explore where dyadic status effects mostly emerge from in social groups. Specifically, we consider whether some dyads generate greater dyadic status effects than others.

Status evaluations are private and unspoken. In spite of this, individuals are skilled at inferring the status that they receive from peers. This latter work has shown that social groups punish individuals through exclusion when they behave in ways that suggest that they do not

know their place (Anderson et al., 2006, 2008). Similarly, if a target misconstrues the status they receive from one of their dyadic peers, this could lead to strained relationships and punishment within the specific dyad. While unspoken, status evaluations are observable through status-relevant behaviors such as deference. Many of these behaviors take place at the dyadic level, suggesting that targets infer the status conferred on them by a given perceiver.

We suggest that an important driver of dyadic status effects is the status distance between perceivers and targets in a given dyad. Status distance is the magnitude of status differences between two individuals (Doyle et al., 2016). It is separate from status differences, which take the direction of the difference into account (i.e., whether one person has higher or lower status than the other). Instead, status distance is solely determined by the absolute difference between two individuals' status in the group's consensus. We suggest that as status distance increases in dyads, each member generates larger dyadic status effects when evaluating the other. Conversely, as status distance decreases, we suggest that dyadic status effects likely decrease. We theorize that this pattern emerges because generating dyadic status effects may stimulate conflict between the two members of the dyad—particularly when they are at a close status distance—and it may precipitate changes in the hierarchy where one member of the dyad usurps the other—again, particularly when they are at a close status distance.

We distinguish between negative dyadic status effects and positive dyadic status effects. Negative dyadic status effects emerge when a target is perceived as especially low status by a particular perceiver relative to the group's consensus. In contrast, positive dyadic status effects reflect the converse—a target is perceived as especially high status by a perceiver relative to consensus. We suggest that negative dyadic status effects are communicated through reduced deference, diminished influence, and other behaviors that signal that the perceiver does not fully

support the target's position in the status hierarchy—potentially provoking status conflict that challenges or disrupts the status order (Bendersky & Hays, 2017). In contrast, we suggest that positive dyadic status effects are communicated through behaviors that signal that the perceiver supports the target attaining a higher position in the status hierarchy, particularly through status-sharing behaviors, in which one person uses their status to boost the status of another person (Hoff et al., 2025).

We propose that individuals are less likely to generate negative dyadic status effects toward peers that occupy nearby positions in the status hierarchy. These individuals experience greater status rivalry, because each represents a plausible competitor for the other's standing (Kilduff et al., 2010). Consequently, a negative dyadic status effect that deviates from the group's consensus and manifests in an observable behavior by the target is likely to be interpreted as a status challenge when it comes from someone of similar status standing. In contrast, the same deviation is less likely to threaten the target when it originates from someone whose status is substantially higher or lower, because the two individuals are less direct competitors in the status hierarchy.

Accordingly, we expect perceivers to exercise greater restraint when evaluating peers who occupy similar positions in the hierarchy. Generating negative dyadic status effects toward these nearby peers may unnecessarily initiate status conflict, proving costly. By contrast, these perceivers have greater latitude to form idiosyncratic negative status evaluations of individuals whose status is much higher or lower than their own, because these judgments are less likely to precipitate status conflict. As a result, we predict that negative dyadic status effects increase in absolute magnitude—that is, become more negative—as status distance increases.

Although status research has focused on the benefits that emerge from targets achieving high levels of status at the consensus-level in social groups, we expect that there could also be benefits from receiving positive dyadic status effects. In other words, as a target, being perceived as especially high status by a perceiver should generate benefits. For instance, positive dyadic status effects may mean that the target is more likely to receive deference from that particular perceiver (Joshi & Knight, 2015). Perhaps even more saliently, this may lead to status-sharing behaviors—where perceivers boost the status of a peer. Over time, this may enable the target to generate greater overall status in the social group. Thus, when a member of a social group confers positive dyadic status effects on others, they may provide them with benefits that could lead these targets to increase their status in future. This represents an adverse situation for the perceiver when the two occupy similar positions in the status hierarchy, because the aforementioned benefits may enable the target to increase their status at the perceiver's expense. Thus, we suggest that members of social groups will tend to shy away from conferring positive dyadic status effects on those close to them in the hierarchy, to avoid losing their own status to them. Together, whether dyadic status effects are positive or negative, we predict:

Hypothesis 1. There is a positive linear relationship between status distance within a dyad and the magnitude, or absolute value, of dyadic status effects.

Method

Transparency and Openness

Our study was not preregistered. We report all measures and exclusions, and we follow Journal Article Reporting Standards throughout (Kazak, 2018). Anonymized data and analysis

code are available on the Open Science Framework

(https://osf.io/9tgw4/overview?view_only=bbc5d92ed492446ebe7afad3cfed6e08). We analyzed data using R (version 4.5.1; R Core Team, 2025) and the lme4 (Version 1.1-37; Bates et al., 2015), lmerTest (version 3.1-3; Kuznetsova et al., 2017) and TripleR packages (version 1.5.4; Schönbrodt et al., 2012). We also used the ggplot2 package to produce graphics for this manuscript (version 4.0.0; Wickham, 2016). We used ChatGPT-5.6 Sol (OpenAI, 2026) to improve code for data analysis and to revise selected sentences in this manuscript prior to submission. The authors reviewed and verified all generated output and take responsibility for the final analyses and text.

Participants and Procedure

We examined status evaluations and, specifically, the component of status that emerges through dyadic status effects, by analyzing peer evaluations of status among individuals working together from three separate samples, using diverse pools of participants and measures of status ($N = 12,464$ status observations). A status observation is one directed perceiver-to-target evaluation, where we leverage multiple items as repeated indicators of the same evaluation to separate dyadic status effects from random error. In the first two samples, sample sizes were determined by the number of participants in the programs that we chose, restricted to those who answered our surveys. In the third sample, we reanalyzed data that were previously published by Yu and Kilduff (2020) with these authors' permission. We conducted SRM analyses (Kenny & La Voie, 1984), widely used to partition variance in interpersonal perceptions, which estimate the proportions of variance attributable to individual (i.e., perceiver and target) and dyadic effects.

Sample 1: We issued surveys¹ to 347 students enrolled in a Master of Business Administration (MBA) program at a European business school. Students were designated to 40 groups of eight or nine for the first year of the program to complete assignments. 325 participants completed the survey (61.4% men, 38.6% women, $M_{age} = 30.59$) and rated each peer in the group, excluding themselves, on the statements “I respect this person” and “I admire this person” (1 = *strongly disagree*, 7 = *strongly agree*; $r = .57, p < .001$).

Sample 2: We issued surveys to 1,029 students enrolled in an Introduction to Organizational Behavior course at a Midwestern university in the United States. Students were assigned to 149 groups of five to eight to complete a variety of tasks. 652 participants completed the survey (58.3% men, 41.7% women, $M_{age} = 19.75$). Status evaluations were collected by asking participants to rate how much “status”, “respect” and “admiration” each of their peers had relative to others in the group ($\alpha = .82$).

Sample 3: Participants were 244 students enrolled at an engineering school in eastern China (54.1% men, 45.9% women). Students were randomly assigned to one of eight cohorts of 26-33 for the entire undergraduate program. Status evaluations were collected during the second semester by asking participants to rate their agreement with the following statements concerning each other individual in their cohort: “has high status; i.e., is earning the respect and admiration of the other cohort members” and “has a lot of influence over others’ thoughts and decisions” (1 = *not at all*, 7 = *a lot*) ($r = .74, p < .001$). These data were originally reported by Yu and Kilduff (2020) and are reanalyzed here with the authors’ permission.

¹ Survey data were collected as part of wider initiatives: a full list of survey questions is included in *Supplementary Online Materials*

Results

Status evaluations can be driven by variance that emerges at three levels: (1) *perceiver effects*, where some individuals confer higher or lower status than others across targets, (2) *target effects*, where some individuals are conferred higher or lower status than others across perceivers, and (3) *dyadic effects*, reflecting ratings that are uniquely high or low for a particular perceiver–target pair after accounting for perceiver and target effects.²

Perceiver and target effects represent the individual variance components, while dyadic effects represent the dyadic variance component. Although past scholars have used SRM to analyze status evaluations (e.g., DesJardins et al., 2015), these prior models appear to have confounded random error with dyadic status effects, since SRM requires multiple items from the same evaluation to separate random error (Schönbrodt et al., 2012). This is an especially important consideration when estimating dyadic effects, since they are less stable than target or perceiver effects (Kenny, 2019). Thus, to the best of our knowledge and based on an extensive literature review, our results may be among the first to evaluate the relative importance of dyadic status effects in status conferral.

SRM Results. In *Table 1*, we present an SRM analysis of status evaluations for our three samples. In the first sample, we found that 18.7% (*variance estimate* = 0.20, $p < .001$) of variance emerged at the perceiver level. Much of this variance may emerge because participants respond to surveys differently (e.g., some tend to use the upper end of scales, and others the lower end). Thus, we suspect that much of the perceiver variance in this sample, and our other samples, reflects this tendency rather than underlying differences in status conferral, which are better captured in estimates of target variance and dyadic variance. To this end, 9.3% (*variance*

² SRM scholars sometimes refer to the perceiver effect as the actor effect, the target effect as the partner effect, and the dyadic effect as the relationship effect; these terms are used interchangeably.

estimate = 0.10, $p < .001$) of variance emerged at the target level, and 24.2% (*variance estimate* = 0.26, $p < .001$) emerged at the dyadic level, with the remainder attributable to random error. In our second sample, we found that 19.8% (*variance estimate* = 0.26, $p < .001$) of variance emerged at the perceiver level, 19.7% (*variance estimate* = 0.26, $p < .001$) emerged at the target level, and 19.3% (*variance estimate* = 0.25, $p < .001$) emerged at the dyadic level, with the remainder attributable to random error. In our third sample, we found that 51.6% (*variance estimate* = 0.96, $p < .001$) of variance emerged at the perceiver level, 7.4% (*variance estimate* = 0.14, $p < .001$) emerged at the target level, and 14.4% (*variance estimate* = 0.27, $p < .001$) emerged at the dyadic level, with the remainder attributable to random error. Thus, in each of these three samples, a statistically significant amount of variance in status evaluations was driven by dyadic status effects. In two of these samples (Samples 1 and 3) the amount of variance explained by dyadic status effects exceeded the amount explained by target effects, and in one sample (Sample 2) these amounts were comparable.

Table 1

Social Relations Models for Status

Variance Component	Sample 1				Sample 2				Sample 3			
	Estimate	SE	p	%	Estimate	SE	p	%	Estimate	SE	p	%
Perceiver	0.20***	0.04	< .001	18.7	0.26***	0.03	< .001	19.8	0.96***	0.10	< .001	51.6
Target	0.10***	0.02	< .001	9.3	0.26***	0.03	< .001	19.7	0.14***	0.02	< .001	7.4
Dyadic	0.26***	0.02	< .001	24.2	0.25***	0.02	< .001	19.3	0.27***	0.01	< .001	14.4
Random Error	0.52	-	-	47.7	0.54	-	-	41.2	0.49	-	-	26.6
Reciprocity and Reliability												
Generalized Reciprocity	0.03	0.02	.185		-0.01	0.02	.787		0.02	0.03	.532	
Dyadic Reciprocity	0.05**	0.02	.002		0.02	0.02	.314		0.03***	0.01	< .001	
Perceiver Effect Reliability	0.51				0.59				0.88			
Target Effect Reliability	0.61				0.65				0.91			
Dyadic Effect Reliability	0.72				0.63				0.68			
Descriptive Statistics												
Number of Observations	2,324				2,890				7,250			
Number of Groups	40				124				8			
Number of Perceivers	325				652				244			
Average Group Size	8.13				5.26				30.50			

Note. Percentages may not sum to 100% because of rounding. Within-group significance testing from Lashley and Bond (1997).

*** $p < .001$. ** $p < .01$.

Hypothesis Testing. Next, we tested Hypothesis 1. Using SRM, we estimated target status effects for each participant in our samples. Because target effects are deviations from the group mean these scores are centered at zero within each group. In our study, higher target status effects indicate that these individuals were conferred higher status (on average) than others from the same group. Using these target status effects, we were able to estimate status distance within every dyad—that is, the absolute difference between these two members’ target status effects. Accordingly, higher status distance scores indicated that perceivers and targets in a given dyad held positions farther apart (as opposed to closer together) in the group’s status hierarchy. Finally, we included both the perceiver and the target’s target status effect as covariates in our regression models. Including these terms as covariates allowed us to isolate the association between dyadic status effects and status distance. No standardization was applied.

Next, we estimated dyadic status effects for each status observation. The estimated dyadic status effect was calculated by subtracting the estimated target and perceiver effects from group-centered status evaluations. Higher dyadic status effects indicate that a perceiver saw the target more positively than would be expected based on the perceiver effect and the target effect. Using SRM, we calculated these dyadic status effects in every status observation. In our hypothesizing, we were primarily interested in the emergence of dyadic status effects, rather than whether those dyadic status effects were positive or negative, and we therefore used the absolute value of the dyadic status effect as the dependent variable in the test of Hypothesis 1.

Summary statistics are presented in *Table 2*.

Table 2

Descriptive Statistics & Intercorrelations Among Study Variables

Sample 1

Variable	M	SD	1	2	3
1 Status Distance	0.38	0.43			
2 Dyadic Status Effect	0.00	0.52	0.00		
3 (Perceiver) Target Status	0.00	0.38	-0.32***	0.00	
4 (Target) Target Status	0.00	0.38	-0.32***	0.00	-0.14***

Sample 2

Variable	M	SD	1	2	3
1 Status Distance	0.68	0.59			
2 Dyadic Status Effect	0.00	0.49	-0.10***		
3 (Perceiver) Target Status	0.00	0.57	-0.03	0.00	
4 (Target) Target Status	0.00	0.57	-0.03	0.00	-0.22***

Sample 3

Variable	M	SD	1	2	3
1 Status Distance	0.40	0.37			
2 Dyadic Status Effect	0.00	0.61	-0.03*		
3 (Perceiver) Target Status	0.00	0.38	0.30***	0.00	
4 (Target) Target Status	0.00	0.38	0.30***	0.00	-0.03**

Note. *** $p < .001$. ** $p < .01$. * $p < .05$.

In Hypothesis 1, we predicted that the status distance between members of a dyad would be positively related to the magnitude of dyadic status effects that emerged. We estimated linear mixed-effects models with random intercepts for groups and dyads, thereby accounting for non-independence of observations and allowing the average magnitude of dyadic status effects to vary across groups and dyads. We found supporting evidence for our hypothesis in all three of our samples, where in each case there was a significant positive association between status distance and the magnitude of dyadic status effects (Sample 1: $b = 0.16$, $t(2,018) = 6.73$, $p < .001$; Sample 2: $b = 0.05$, $t(2,681) = 4.38$, $p < .001$; Sample 3: $b = 0.20$, $t(3,555) = 12.76$, $p < .001$; see *Table 3* and *Figure 1*).

Table 3

Regressing Magnitude of Dyadic Status Effects on Status Distance

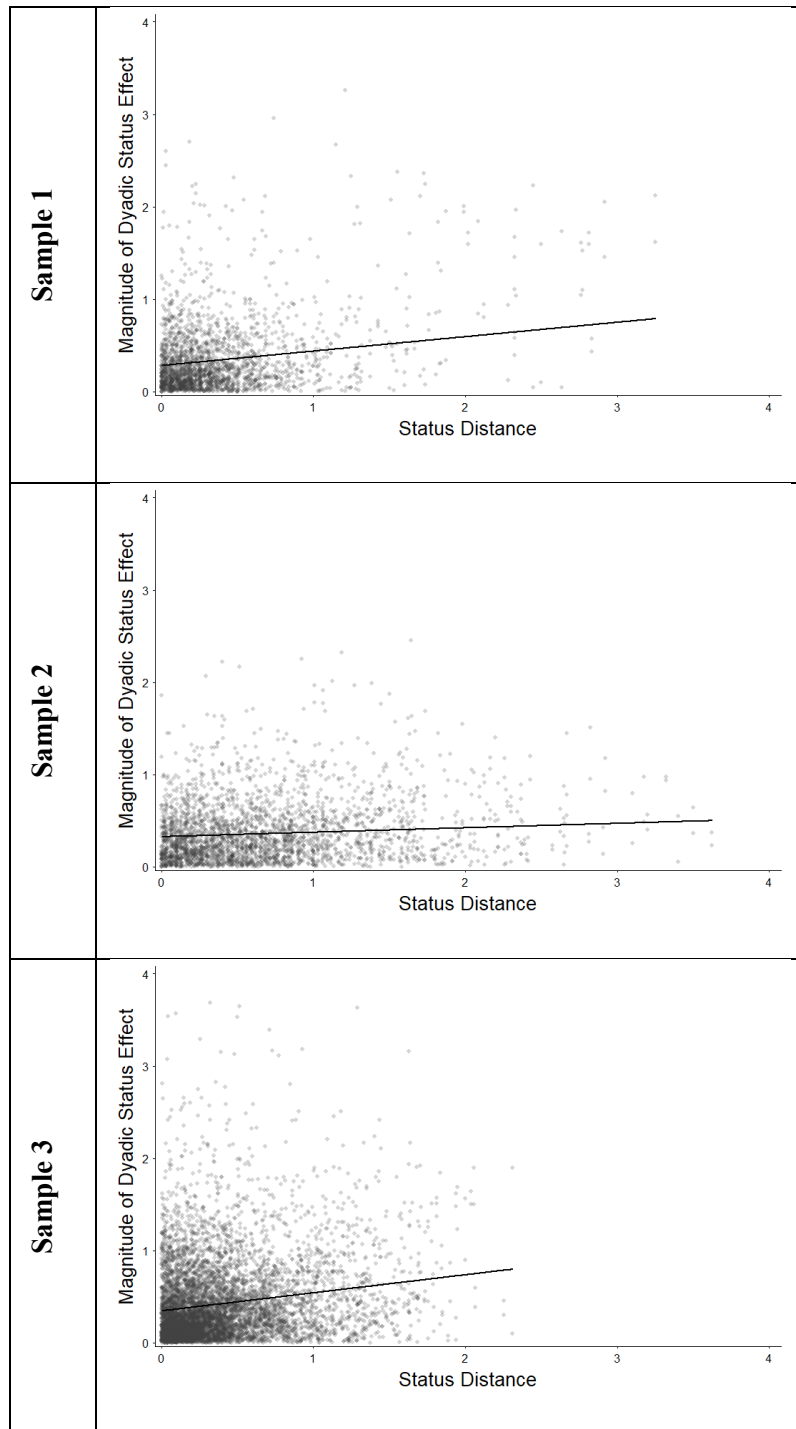
	Sample 1				Sample 2				Sample 3			
Fixed Effects	b	SE	t (df)	p	b	SE	t (df)	p	b	SE	t (df)	p
Intercept	0.28***	0.02	12.66 (49)	< .001	0.32***	0.01	23.78 (2,124)	< .001	0.34***	0.02	18.58 (9)	< .001
(Perceiver) Target Status	-0.01	0.02	-0.38 (2,318)	.701	0.01	0.01	0.74 (2,759)	.461	-0.07***	0.01	-5.11 (7,065)	< .001
(Target) Target Status	-0.18***	0.02	-8.52 (2,318)	< .001	-0.03**	0.01	-3.01 (2,759)	.003	0.10***	0.01	7.55 (7,065)	< .001
Status Distance	0.16***	0.02	6.73 (2,018)	< .001	0.05***	0.01	4.38 (2,681)	< .001	0.20***	0.02	12.76 (3,555)	< .001
Random Effects												
Group	0.00				0.00				0.05			
Dyadic	0.12				0.11				0.05			
Residual	0.35				0.31				0.42			
Descriptive Statistics												
Number of Observations	2,324				2,890				7,250			
Number of Groups	40				124				8			
Number of Dyads	1,162				1,445				3,625			

Note. (Perceiver) target status and (target) target status are the group-centered SRM target effects of the individual giving and receiving each status evaluation, respectively. Status distance is the absolute difference between these two scores. Entries are unstandardized regression coefficients. Models included random intercepts for groups and dyads. p-values were calculated using Satterthwaite's approximation for the degrees of freedom.

***p < .001. **p < .01.

Figure 1

Effect of Status Distance on Magnitude of Dyadic Status Effects



Note. The y-axis shows the absolute value of dyadic status effects. Solid lines show predictions from the mixed-effects models in Table 2, with perceiver and target status held at zero.

As a supplementary analysis, we considered whether status distance was associated with negative or positive dyadic status effects, in particular. We reran our models, but this time included an interaction term using dummy-coding that distinguished negative from positive dyadic status effects. Results showed that these interactions were not statistically significant in two samples (Sample 1: $b = -0.04$, $t(2,282) = -1.12$, $p = .263$; Sample 3: $b = 0.01$, $t(7,240) = 0.42$, $p = .671$; see *Table 4*). Simple slope analyses confirmed that there was a positive association between status distance and negative dyadic status effects in both samples (Sample 1: $b = 0.17$, $p < .001$; Sample 3: $b = 0.19$, $p < .001$), as well as a positive association between status distance and positive dyadic status effects in both samples (Sample 1: $b = 0.13$, $p < .001$; Sample 3: $b = 0.20$, $p < .001$).

However, this interaction was significantly negative in one sample (Sample 2: $b = -0.07$, $t(2,823) = -3.28$, $p = .001$). Simple slope analyses confirmed that dyads at greater status distance in this sample conferred greater negative dyadic status effects on one another ($b = 0.08$, $p < .001$), but not greater positive dyadic status effects ($b = 0.02$, $p = .305$).

Table 4
Regressing Dyadic Status Effects on Status Distance

	Sample 1				Sample 2				Sample 3			
Fixed Effects	b	SE	t (df)	p	b	SE	t	p	b	SE	t	p
Intercept	0.29***	0.02	11.93 (73)	< .001	0.29***	0.02	17.76 (422)	< .001	0.34***	0.02	16.95 (12)	< .001
(Perceiver) Target Status	-0.01	0.02	-0.35 (2,316)	.725	0.01	0.01	1.02 (2,757)	.306	-0.07***	0.01	-5.16 (7,065)	< .001
(Target) Target Status	-0.19***	0.02	-8.84 (2,316)	< .001	-0.03**	0.01	-2.96 (2,757)	.003	0.11***	0.01	7.69 (7,065)	< .001
Status Distance	0.17***	0.03	5.88 (2,220)	< .001	0.08***	0.01	5.60 (2,838)	< .001	0.19***	0.02	9.62 (5,287)	< .001
Dyadic Status Effect Direction	-0.02	0.02	-1.16 (2,283)	.246	0.07***	0.02	3.88 (2,805)	< .001	0.01	0.01	0.97 (7,219)	.333
Status Distance × Dyadic Status Effect Direction	-0.04	0.03	-1.12 (2,282)	.263	-0.07**	0.02	-3.28 (2,823)	.001	0.01	0.03	0.42 (7,240)	.671
Random Effects												
Group	0.00				0.00				0.05			
Dyadic	0.12				0.11				0.05			
Residual	0.35				0.31				0.42			
Descriptive Statistics												
Number of Observations	2,324				2,890				7,250			
Number of Groups	40				124				8			
Number of Dyads	1,162				1,445				3,625			

Note. (Perceiver) target status and (target) target status are the group-centered SRM target effects of the individual giving and receiving each status rating, respectively. Status distance is the absolute difference between these two scores. Entries are unstandardized regression coefficients. Models included random intercepts for groups and dyads. p-values were calculated using Satterthwaite's approximation for the degrees of freedom. Dyadic status effect direction is dummy-coded 1 for positive dyadic status effects and 0 for negative dyadic status effects. Dyadic effects equal to exactly zero were coded as positive. ***p < .001. **p < .01.

Discussion

Given the vital role of status in social interaction, scholars have unsurprisingly devoted considerable attention to understanding its drivers. This work has documented numerous individual-level attributes and behaviors that confer status advantages on the individuals who possess or display them—through the consensus that emerges in social groups.

However, nascent research has highlighted that while groups reach *some* measure of consensus over status, individuals within these groups still frequently disagree with one another over their own status positions (Kilduff et al., 2016), perceive the shape of the hierarchy differently (Tan & Rider, 2023), and simply misrepresent the status hierarchy (Yu et al., 2023), meaning that considerable dissensus also emerges. In the present research, we were interested in understanding how much variance in status evaluations emerges through dyadic status effects—one reason for dissensus—and advancing theory as to where these effects emerge from.

Across three empirical samples involving groups of diverse sizes, working on different tasks in different geographical regions ranging from the United States to Europe and China ($N = 12,464$ status observations), we found that a statistically significant amount of variance in status evaluations emerged from dyadic status effects. In two of our three samples, we found that the estimated variance attributable to dyadic status effects exceeded that attributable to target effects.

We then looked at *where* these dyadic status effects emerged from, i.e., which dyads in these groups produced greater dyadic status effects than others. We focused our theorizing and empirical examination on how the status distance between perceivers and targets shapes the generation of dyadic status effects. We found in all three samples that dyadic status effects were larger in magnitude when members of a dyad were located farther apart in the status hierarchy

(i.e., at greater status distances). In contrast, when status distance was lower, dyadic status effects decreased.

Theoretical Contributions

We make a number of important contributions to the status literature through the present research. First, we contribute to a deeper understanding of status by highlighting its multilevel and dyadic foundations. We draw attention to the importance of dyadic status effects—and thereby position these as an important and understudied category of status antecedents in the literature. Prior work has identified numerous antecedents to status conferral, with a focus on attributes that widely signal social worth and competence (Bunderson, 2003; Correll & Ridgeway, 2006), behaviors such as overconfidence, expressions of authentic pride, and humor (Anderson et al., 2012; Bitterly et al., 2017; Cheng et al., 2010), as well as prosocial acts that are valued in groups, including generosity, moral virtue, and altruism (Bai et al., 2020; Flynn, 2003; Hardy & Van Vugt, 2006; Willer, 2009). However, these antecedents of status have been studied as individual-level effects, i.e., linking a particular attribute or behavior with the status a target is granted across different social groups. In the present research, we highlight that dyadic status effects also matter greatly for status conferral, arguing that a shift in focus is required to deepen collective understanding of status emergence. Specifically, we point out that assessing another individual's competence is highly subjective and often shaped by dyadic forces, which leads to dyadic status effects. This is an important contribution to the literature that advances another lens for studying status antecedents.

Second, we identify status distance as a robust correlate of dyadic-effect magnitude. Specifically, we argued that dyadic status effects are larger in dyads where members are located at more distant (as opposed to more similar) positions in the status hierarchy—which was

supported in all three of the samples in our study. Thus, beyond highlighting the importance of dyadic status effects in the context of status conferral, we contribute novel explanations about where these effects are most likely to emerge from in social groups.

Third, the present research identifies a highly important statistical component of status dissensus (Anderson & Brown, 2010). Recent work has demonstrated that individuals perceive status hierarchies differently (Yu & Kilduff, 2020), which partly reflects dispositional variation in the ability to accurately infer the status consensus (Yu et al., 2023). Those with high status tend to see greater steepness in the hierarchy (Tan & Rider, 2023). And moreover, individuals often disagree with one another over their own status positions (Kilduff et al., 2016). Building on this work, the present research demonstrates that a significant component of status evaluations is specific to dyadic relationships. This is an important insight, because it highlights one reason why status is not completely agreed upon, i.e., reflected in the consensus. All else equal, greater dyadic status effects reduce consensus over status. Given continued scholarly focus on the effects of hierarchy on performance (e.g., Anderson & Brown 2010; Greer et al. 2018), our work highlights the critical need to understand whether dyadic status effects have downstream effects on performance.

Practical Implications

Our findings have several practical implications. Prior scholarship has suggested that a lack of consensus over status hinders performance, for instance by lowering cohesion and fostering status conflict; and this is especially true in more complex task environments where collaboration and coordination are essential (Bunderson et al., 2016; Yu et al., 2023). Our findings identify one component of dissensus—dyadic-specific variation in status evaluations. Thus, leaders may wish to intervene in ways that encourage greater consensus by reducing

dyadic status effects. One intervention for future field testing may be to clarify evaluation criteria and disseminate relevant performance information throughout the group. In doing so, leaders may be able to reduce dyadic status effects, potentially increasing consensus over status and unlocking the associated benefits. Interventions may warrant focusing on dyads characterized by greater status distance, since we found that dyadic status effects were largest in these pairs.

Leaders should be alert to potential bias in their own status evaluations toward subordinates, particularly when these judgments affect task delegation, performance evaluations, and promotions. Although we did not measure leaders' status evaluations of subordinates in our study, since our groups did not have a formal hierarchy, we expect that leaders are far from immune to idiosyncratic status evaluations, and that the consequences are likely to be severe.

Limitations and Future Directions

Our findings may not generalize beyond student groups, and the specific tasks participants undertook in our samples. Additionally, while the self-managed groups in each of our samples allowed us to hold formal rank differences constant, the associated dynamics do not fully reflect structures in many teams, for instance many workplace teams in hierarchical organizations. Future research should replicate our analyses in workplace teams, and other naturalistic groups with formal rank differences.

An extensive literature has demonstrated that status hierarchies form rapidly and are self-reinforcing in groups, whereby individuals who attain status can use its associated advantages to preserve and even boost their positions (Berger et al., 1972; Merton, 1968). However, a burgeoning literature has started to consider the effects of status dynamics, including status momentum (Pettit et al., 2013) and trajectories (Pettit et al., 2013; Pettit & Marr, 2020), which highlights the fluidity of status. In keeping with this work, because we measured status at one

time point, we could not determine whether dyadic effects may change over time (e.g., get bigger, become more numerous). Longitudinal designs would better capture these dynamics and potentially reveal additional theoretical insights.

Another limitation of our work is that we were unable to test the psychological mechanisms associated with the emergence of dyadic status effects in our samples. Although the observed association between status distance and dyadic status effects was consistent with our theory, the proposed psychological mechanisms remain untested. This means that there are alternative psychological mechanisms that may explain the same pattern in the data. It is important that future research test these proposed mechanisms and develop alternative explanations.

We advocate for more work to understand when dyadic status effects are more likely to impact status in social groups. In the present research, we identified status distance between dyadic members as a potential driver; however, this mechanism best explains *where* dyadic status effects emerge in a given social group, rather than *what* drives dyadic status effects across groups, which would require further studies using different designs.

Finally, we highlight the need for research on downstream consequences of dyadic status effects. Research to date has considered how an individual's consensus-level status shapes their behavior, as well as other individuals' behaviors toward them. We foresee considerable potential in exploring the downstream behavioral outcomes of dyadic status effects. We suggest particular attention be paid to dyadic behaviors; e.g., influence, deference (Joshi & Knight, 2015), status-sharing (Hoff et al., 2025) and status conflict (Bendersky & Hays, 2012). We also advocate for work that considers how dyadic status effects shape downstream outcomes for groups, particularly performance. Much work has considered the extent to which status hierarchies are

functional—that is, whether they enhance performance (Greer et al., 2018), with a critical contingency theorized to be whether the most competent and prosocially motivated individuals emerge with the highest status at the consensus-level (Anderson & Brown, 2010). Moreover, research has shown that participants' mental representations of hierarchy have profound implications for performance, particularly by stimulating status conflict (Yu et al., 2019, 2023). We suggest that future research examines the effects of dyadic status effects on behaviors and performance to help enhance collective understanding of the factors that drive more or less functional status hierarchies.

Conclusion

Status hierarchies are universal and have profound effects on individuals and groups. Prior research has examined the antecedents of status evaluations, but has largely focused on individual-level effects, i.e., understanding ways in which targets' attributes and behaviors affect the status conferred on them across social groups. However, we highlight that a substantial portion of variation in status evaluations is specific to dyads rather than attributable to group consensus. Thus, we provide a fresh perspective on the drivers of status and highlight the need for more research into this important topic for status scholars. We encourage researchers to build on our insights and deepen collective knowledge of the antecedents of status, and the consequences of dyadic status effects.

References

- Anderson, C., Ames, D. R., & Gosling, S. D. (2008). Punishing hubris: The perils of overestimating one's status in a group. *Personality and Social Psychology Bulletin*, 34(1), 90–101. <https://doi.org/10.1177/0146167207307489>
- Anderson, C., Brion, S., Moore, D. A., & Kennedy, J. A. (2012). A status-enhancement account of overconfidence. *Journal of Personality and Social Psychology*, 103(4), 718–735. <https://doi.org/10.1037/a0029395>
- Anderson, C., & Brown, C. E. (2010). The functions and dysfunctions of hierarchy. *Research in Organizational Behavior*, 30, 55–89. <https://doi.org/10.1016/j.riob.2010.08.002>
- Anderson, C., Hildreth, J. A. D., & Howland, L. (2015). Is the desire for status a fundamental human motive? A review of the empirical literature. *Psychological Bulletin*, 141(3), 574–601. <https://doi.org/10.1037/a0038781>
- Anderson, C., John, O. P., Keltner, D., & Kring, A. M. (2001). Who attains social status? Effects of personality and physical attractiveness in social groups. *Journal of Personality and Social Psychology*, 81(1), 116–132. <https://doi.org/10.1037/0022-3514.81.1.116>
- Anderson, C., Srivastava, S., Beer, J. S., Spataro, S. E., & Chatman, J. A. (2006). Knowing your place: Self-perceptions of status in face-to-face groups. *Journal of Personality and Social Psychology*, 91(6), 1094–1110. <https://doi.org/10.1037/0022-3514.91.6.1094>
- Bai, F., Ho, G. C. C., & Yan, J. (2020). Does Virtue Lead to Status? Testing the Moral Virtue Theory of Status Attainment. *Journal of Personality and Social Psychology*, 118(3), 501–531. <https://doi.org/10.1037/pspi0000192>
- Bates, D., Mächler, M., Bolker, B., & Walker, S. (2015). Fitting Linear Mixed-Effects Models Using lme4. *Journal of Statistical Software*, 67, 1–48. <https://doi.org/10.18637/jss.v067.i01>

- Bendersky, C., & Hays, N. A. (2012). Status conflict in groups. *Organization Science*, 23(2), 323–340. <https://doi.org/10.1287/orsc.1110.0734>
- Bendersky, C., & Hays, N. A. (2017). The Positive Effects of Status Conflicts in Teams Where Members Perceive Status Hierarchies Differently. *Social Psychological and Personality Science*, 8(2), 124–132. <https://doi.org/10.1177/1948550616667614>
- Berger, J., Cohen, B. P., & Zelditch, M. (1972). Status Characteristics and Social Interaction. *American Sociological Review*, 37(3), 241–255. <https://doi.org/10.2307/2093465>
- Bitterly, T. B., Schweitzer, M. E., & Brooks, A. W. (2017). Risky business: When humor increases and decreases status. *Journal of Personality and Social Psychology*, 112(3), 431–455. <https://doi.org/10.1037/pspi0000079>
- Blau, P. M. (1964). *Exchange and power in social life*. J Wiley & Sons.
- Bunderson, J. S. (2003). Recognizing and utilizing expertise in work groups: A status characteristics perspective. *Administrative Science Quarterly*, 48(4), 557–591. <https://doi.org/10.2307/3556637>
- Bunderson, J. S., Van Der Vegt, G. S., Cantimur, Y., & Rink, F. (2016). Different views of hierarchy and why they matter: Hierarchy as inequality or as cascading influence. *Academy of Management Journal*, 59(4), 1265–1289. <https://doi.org/10.5465/amj.2014.0601>
- Cheng, J. T., Tracy, J. L., & Henrich, J. (2010). Pride, personality, and the evolutionary foundations of human social status. *Evolution and Human Behavior*, 31(5), 334–347. <https://doi.org/10.1016/j.evolhumbehav.2010.02.004>
- Correll, S. J., & Ridgeway, C. L. (2006). Expectation States Theory. In J. Delamater (Ed.), *Handbook of Social Psychology. Handbooks of Sociology and Social Research* (pp. 29–51). Springer, Boston, MA. https://doi.org/10.1007/0-387-36921-X_2

- de Jong, S. B., Van der Vegt, G. S., & Molleman, E. (2007). The Relationships Among Asymmetry in Task Dependence, Perceived Helping Behavior, and Trust. *Journal of Applied Psychology*, 92(6), 1625–1637. <https://doi.org/10.1037/0021-9010.92.6.1625>
- DesJardins, N. M. L., Srivastava, S., Küfner, A. C. P., & Back, M. D. (2015). Who Attains Status? Similarities and Differences Across Social Contexts. *Social Psychological and Personality Science*, 6(6), 692–700. <https://doi.org/10.1177/1948550615580171>
- Doyle, S. P., Lount, R. B., Wilk, S. L., & Pettit, N. C. (2016). Helping Others Most When They Are Not Too Close: Status Distance as a Determinant of Interpersonal Helping in Organizations. *Academy of Management Discoveries*, 2(2), 155–174. <https://doi.org/10.5465/amd.2014.0104>
- Flynn, F. J. (2003). How Much Should I Give and How Often? The Effects of Generosity and Frequency of Favor Exchange on Social Status and Productivity. *Academy of Management Journal*, 46(5), 539–553. <https://doi.org/10.2307/30040648>
- Foschi, M. (2000). Double Standards for Competence: Theory and Research. *Annual Review of Sociology*, 26, 21–42. <https://doi.org/10.1146/annurev.soc.26.1.21>
- Greer, L. L., de Jong, B. A., Schouten, M. E., & Dannals, J. E. (2018). Why and when hierarchy impacts team effectiveness: A meta-analytic integration. *Journal of Applied Psychology*, 103(6), 591–613. <https://doi.org/10.1037/apl0000291>
- Grosz, M. P., van Aert, R. C. M., & Back, M. D. (2024). A Meta-Analytic Review of the Associations of Personality, Intelligence, and Physical Size with Social Status. *Psychological Bulletin*, 150(3), 253–283. <https://doi.org/10.1037/bul0000416>

- Hardy, C. L., & Van Vugt, M. (2006). Nice guys finish first: The competitive altruism hypothesis. *Personality and Social Psychology Bulletin*, 32(10), 1402–1413.
<https://doi.org/10.1177/0146167206291006>
- Henrich, J., & Gil-White, F. J. (2001). The evolution of prestige: Freely conferred deference as a mechanism for enhancing the benefits of cultural transmission. *Evolution and Human Behavior*, 22(3), 165–196. [https://doi.org/10.1016/S1090-5138\(00\)00071-4](https://doi.org/10.1016/S1090-5138(00)00071-4)
- Hoff, M., Rucker, D. D., & Galinsky, A. D. (2025). The Vicious Cycle of Status Insecurity. *Journal of Personality and Social Psychology*, 128(1), 101–122.
<https://doi.org/10.1037/pspi0000473>
- Inesi, M. E., & Cable, D. M. (2015). When Accomplishments Come Back to Haunt You: The Negative Effect of Competence Signals on Women's Performance Evaluations. *Personnel Psychology*, 68(3), 615–657. <https://doi.org/10.1111/peps.12083>
- Jones, S. L., & Shah, P. P. (2016). Diagnosing the locus of trust: A temporal perspective for trustor, trustee, and dyadic influences on perceived trustworthiness. *Journal of Applied Psychology*, 101(3), 392–414. <https://doi.org/10.1037/apl0000041>
- Joshi, A., & Knight, A. P. (2015). Who defers to whom and why? Dual pathways linking demographic differences and dyadic deference to team effectiveness. *Academy of Management Journal*, 58(1), 59–84. <https://doi.org/10.5465/amj.2013.0718>
- Judge, T. A., Bono, J. E., Ilies, R., & Gerhardt, M. W. (2002). Personality and leadership: A qualitative and quantitative review. *Journal of Applied Psychology*, 87(4), 765–780.
<https://doi.org/10.1037/0021-9010.87.4.765>
- Kalish, Y., & Luria, G. (2021). Traits and time in leadership emergence: A longitudinal study. *Leadership Quarterly*, 32(2), 101443. <https://doi.org/10.1016/j.leaqua.2020.101443>

- Kazak, A. E. (2018). Editorial: Journal article reporting standards. *American Psychologist*, 73(1), 1–2. <https://doi.org/10.1037/amp0000263>
- Kenny, D. A. (2019). *Interpersonal perception: The foundation of social relationships*. Guilford Publications.
- Kenny, D. A., & La Voie, L. (1984). The social relations model. *Advances in Experimental Social Psychology*, 18, 141–182. [https://doi.org/10.1016/S0065-2601\(08\)60144-6](https://doi.org/10.1016/S0065-2601(08)60144-6)
- Kilduff, G. J., Elfenbein, H. A., & Staw, B. M. (2010). The psychology of rivalry: A relationally dependent analysis of competition. *Academy of Management Journal*, 53(5), 943–969. <https://doi.org/10.5465/AMJ.2010.54533171>
- Kilduff, G. J., Willer, R., & Anderson, C. (2016). Hierarchy and its discontents: Status disagreement leads to withdrawal of contribution and lower group performance. *Organization Science*, 27(2), 373–390. <https://doi.org/10.1287/orsc.2016.1058>
- Kunda, Z. (1990). The Case for Motivated Reasoning. *Psychological Bulletin*, 108(3), 480–498. <https://doi.org/10.1037/0033-2909.108.3.480>
- Kuznetsova, A., Brockhoff, P. B., & Christensen, R. H. B. (2017). lmerTest package: Tests in linear mixed effects models. *Journal of Statistical Software*, 82(13), 1–26. <https://doi.org/10.18637/jss.v082.i13>
- Lashley, B. R., & Bond, C. F. (1997). Significance Testing for Round Robin Data. *Psychological Methods*, 2(3), 278–291. <https://doi.org/10.1037/1082-989X.2.3.278>
- Magee, J. C., & Galinsky, A. D. (2008). Social Hierarchy: the self-reinforcing nature of power and status. *Academy of Management Annals*, 2(1), 351–398. <https://doi.org/10.1080/19416520802211628>

Merton, R. K. (1968). The Matthew Effect in Science. *Science*, 159(3810), 56–63.

<https://doi.org/10.1126/science.159.3810.56>

OpenAI. (2026). *Previewing GPT-5.6 Sol: a next-generation model*.

<https://openai.com/index/previewing-gpt-5-6-sol/>

Pettit, N. C., & Marr, J. C. (2020). A trajectories based perspective on status dynamics. *Current Opinion in Psychology*, 33, 233–237. <https://doi.org/10.1016/j.copsyc.2019.10.001>

Pettit, N. C., Sivanathan, N., Gladstone, E., & Marr, J. C. (2013). Rising Stars and Sinking Ships: Consequences of Status Momentum. *Psychological Science*, 24(8), 1579–1584.

<https://doi.org/10.1177/0956797612473120>

R Core Team. (2025). *R: A language and environment for statistical computing (Version 4.5.1)* [Computer software]. R Foundation for Statistical Computing. <https://www.R-project.org/>

Ridgeway, C. L. (1978). Conformity , Group-Oriented Motivation , and Status Attainment in Small Groups. *Social Psychology*, 41(3), 175–188. <https://doi.org/10.2307/3033555>

Ridgeway, C. L., & Berger, J. (1986). Expectations, Legitimation, and Dominance Behavior in Task Groups. *American Sociological Review*, 51(5), 603–617.

<https://doi.org/10.2307/2095487>

Schönbrodt, F., Back, M. D., & Schmukle, S. C. (2012). TripleR: An R package for social relations analyses based on round-robin designs. *Behavior Research Methods*, 44(2), 455–470. <https://doi.org/10.3758/s13428-011-0150-4>

Tan, D., & Rider, C. I. (2023). How the Basis for Status evaluations Varies with Perceiver Status. *Organization Science*, 34(2), 509–531. <https://doi.org/10.1287/orsc.2022.1582>

Wickham, H. (2016). *ggplot2: Elegant graphics for data analysis*. Springer-Verlag New York. <https://doi.org/10.1002/wics.147>

Willer, R. (2009). Groups reward individual sacrifice: The status solution to the collective action problem. *American Sociological Review*, 74(1), 23–43.

<https://doi.org/10.1177/000312240907400102>

Yu, S., Greer, L. L., Halevy, N., & van Bunderen, L. (2019). On Ladders and Pyramids: Hierarchy's Shape Determines Relationships and Performance in Groups. *Personality and Social Psychology Bulletin*, 45(12), 1717–1733. <https://doi.org/10.1177/0146167219842867>

Yu, S., & Kilduff, G. J. (2020). Knowing Where Others Stand: Accuracy and Performance Effects of Individuals' Perceived Status Hierarchies. *Journal of Personality and Social Psychology*, 119(1), 159–184. <https://doi.org/10.1037/pspi0000216>

Yu, S., Kilduff, G. J., & West, T. (2023). Status Acuity: The Ability to Accurately Perceive Status Hierarchies Reduces Status Conflict and Benefits Group Performance. *Journal of Applied Psychology*, 108(1), 114–137. <https://doi.org/10.1037/apl0001024>